LASSO

Lasso (Least Absolute Shrinkage and Selection Operator) regression is a popular method used for variable selection and regularization in linear regression models. It was introduced by Robert Tibshirani in 1996.

In traditional linear regression, the aim is to find the coefficients of the variables that minimize the sum of squared residuals. However, in Lasso regression, an additional penalty term is added to the objective function. This penalty term is the absolute value of the sum of the coefficients, multiplied by a tuning parameter λ. This penalty term is also known as L1 regularization.

The purpose of this penalty term is to shrink the coefficient estimates towards zero and force some of them to become exactly zero. As a result, some variables are selected and some are excluded from the model, which leads to a simpler and more interpretable model.

The value of the tuning parameter λ controls the amount of shrinkage applied to the coefficients. A larger value of λ leads to more shrinkage, and more coefficients are set to zero. The optimal value of λ can be determined using cross-validation techniques.

Lasso regression has several advantages over traditional linear regression, such as reducing overfitting and improving the interpretability of the model. However, it has some limitations, such as difficulty in handling multicollinearity among the variables, and it may not perform well when the number of variables is much larger than the number of observations.

For additional information on LASSO regression please read the following article from CFI:

LASSO - Overview, Uses, Estimation and Geometry

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